

TIC 394710986

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.683676 d
Epoch (t0) 1628.7689 BTJD
Duration 0.0840 d
Depth 1.938 % (19,383 ppm)
SNR 84.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

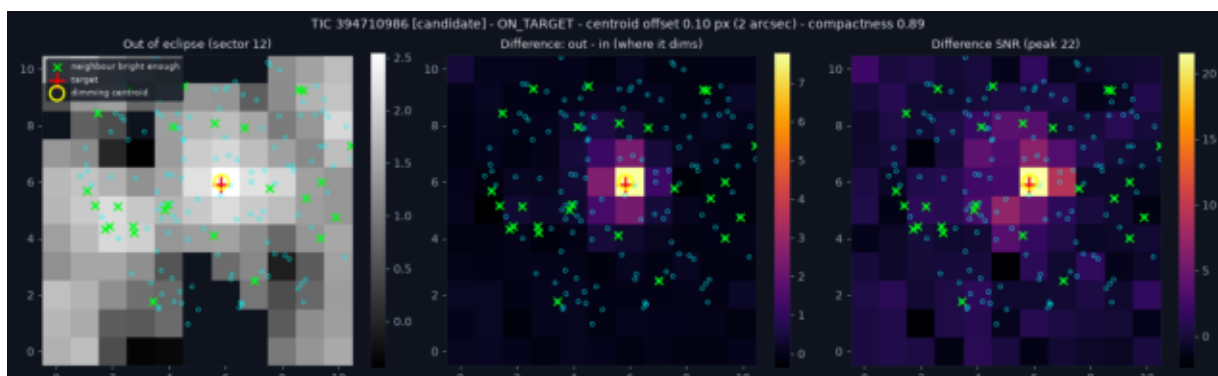
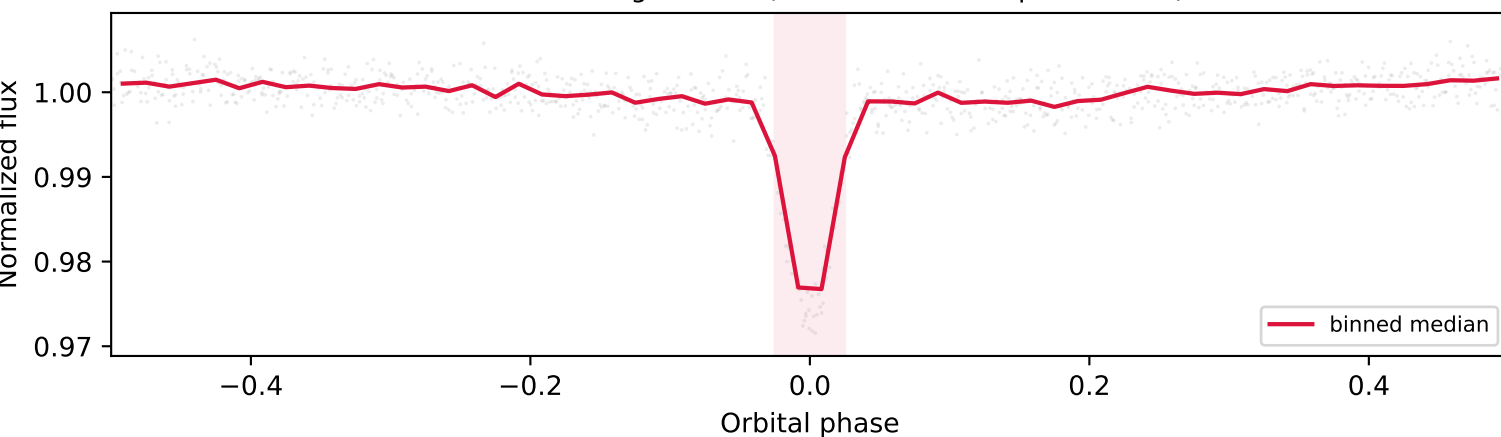
Eclipses observed 14 (gate ≥ 3)
Robust depth ratio 1.07 (gate ≥ 0.50)
Robust depth sigma 64.1 (gate ≥ 7.0)
Odd/even z-score 2.25 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 12 light-curve dip in this sector: 65.7 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 21.62 (gate ≥ 5.0) Compactness 0.89 (gate ≥ 0.50)
Dimming centroid offset 0.10 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 394601747
1.73 px away, dTmag +0.92 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.